VERITY® Syringe Pumps/402 Syringe Pumps

Gilson systems feature a two-way communication interface between the PC and most Gilson instruments. Communication occurs along the Gilson Serial Input/Output Channel (GSIOC).

Each instrument is identified by a unique unit ID, designated by a number between 0 and 63.

402 Syringe Pumps

- The default unit ID of the 402 syringe pumps is 0.

VERITY Syringe Pumps

- The default unit ID of the VERITY® syringe pumps is 11.
- The default baud rate is 115200 (8 bits, even parity, 1 stop bit)
- The baud rate is fixed at 19200 for unit IDs 0–4 and fixed at 57600 for unit IDs 5–7. The baud rate is settable at unit IDs 9–63 using NV-RAM address 7.

Using the PC and software, you:

- specify the instrument you want to control
- issue commands that set operating parameters, control operation, or request information from that instrument

There are two kinds of commands that you can send over the GSIOC:

- **Buffered** commands send instructions to the device. These commands are executed one at a time.
- **Immediate** commands request status information from the device. These commands are executed immediately, temporarily interrupting other commands in progress.

For more information on GSIOC, refer to the GSIOC Technical Manual (part number LT2181).

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GSIOC Command List

In the command list on the following pages, the GSIOC command must be entered in the proper upper or lower case format. If a buffered command requires additional information, there will be italicized text next to the command. The description of the command identifies what to enter in place of the italicized parameter. Also note that if a parameter is optional, it appears within brackets, [ ].

I - Immediate

B - Buffered

VERITY® Syringe Pumps (v1.0.1.5)			402 Syringe Pumps (v2.32)		
Command	*Type*	*Description*	*Command*	*Type*	*Description*
%	I	Requests module identification. Returns character string: "VERITY 4xxx va.b.c.d" where xxx represent the model number and a, b, c, and d represent the firmware version."	%	I	Request module identification Response format: 402SVa.bc where Va.bc is the software version.
$	I	Resets the syringe pump. Returns $ and resets the syringe pump to its power on state.	$	I	Master reset Response format: $ where The returned $ indicates that the machine has been reset in its power-up state.
~n	B	Sets to test mode where: n = 0: Stop any test mode in progress. n = 1: Home machine; run motor sequential pattern test. **Note**: Remove syringe and valve before running this test. n = 9: Reset NV-RAM and initialize to default values. Does not modify serial number, syringe stroke counter, or syringe valve switch counter.			
@	I	Reads non-volatile memory (NV-RAM) at address. Returns "a=v" where: a = the address v = the data for the address Current address is incremented each time the command is sent. Valid addresses are shown in the Non-Volatile Memory Map on page 13.			

VERITY® Syringe Pumps (v1.0.1.5)			402 Syringe Pumps (v2.32)		
Command	*Type*	*Description*	*Command*	*Type*	*Description*
@a[=v]	B	Sets the value at non-volatile memory (NV-RAM) address where: a = the address v (Optional) = the data at the address The current NV-RAM address is set to a. Valid addresses are shown in the Non-Volatile Memory Map on page 13.			
.	I	Reads the most recent '.' command response string. If no previous response, returns '|'. To read the serial number, first send the buffered ".sn" command and then the immediate "." command. Returns ".sn = serial number".			
.str	B	Writes NV-RAM Command (Character) **Note**: Do not exceed the 16 character limit. Verify data using the immediate "." Command.			
?	I	**UNSUPPORTED** Debug Command Response format "ePos/sPos=V0/0 S0/0/0" Numbers are Encoder position/Step position/Target position			

VERITY® Syringe Pumps (v1.0.1.5)			402 Syringe Pumps (v2.32)		
Command	*Type*	*Description*	*Command*	*Type*	*Description*
B	I	Reads the pressure. Returns the value with pressure units.			
B*n*	B	Sets the pressure offset compensation. n = S Sets the zero offset (compensated data) n = Z Clears out the zero offset (uncompensated data) The zero must be cleared before setting a new zero offset.			
D	I	Reads the streaming pressure. Returns the units and then samples separated by colons. Maximum number of samples is 32. Once read, the samples are cleared from the buffer.			
e	I	Reads the current error number and descriptive text. Returns "n", which identifies the error number; see page 15 for a listing of errors. If no error has occurred, returns 0. All errors block buffered commands, except p (home) or e (error).			
e*[n]*	B	Clears error number. n (Optional) Setting a non-zero parameter allows software to raise an error.			

VERITY® Syringe Pumps (v1.0.1.5)			402 Syringe Pumps (v2.32)		
Command	*Type*	*Description*	*Command*	*Type*	*Description*
F	I	Reads syringe size and returns syringe size, minimum flow rate, maximum flow rate, and default flow rate. Returns “syringe size[ul] min-max (default)[mL/min]”			
			F	B	Set syringe motor force Syntax: Fna where n is the syringe identification n = L for left syringe n = R for right syringe. a selects the amplitude of the motor current: 0 = 0% of nominal current (the motor is unpowered) 1 = 25%, 2 = 37.5%, 3 = 50%, 4 = 75%, 5 = 100%. The current level is automatically set according to the syringe type, as follows: 100, 250, 500, 1000 µl: level 3; 5000, 10000, 25000 µl, 39000 steps: level 5. Use this command to set the syringe motor force to a different value if the need arises. Send it after defining the syringe using the B P command, and before to send the first motion command. If a motion is already in progress, this command will only take effect at the next motion. After a delay of about ten seconds of idling, the current will be automatically reduced to level 1 to reduce temperature rise, but it will be restored to the specified value at the beginning of every motion without the need to issue this command each time.
M	I	Returns ‘ab’, where a is the valve motor status. (V)acant , (U)npowered, (P)arked, (R)unning, (E)rror b is the syringe motor status. (U)npowered, (P)arked, (R)unning, (E)rror	M	I	Read syringe status Response format: fnnnnnngmmmmmm– where f is the left syringe status. It can have one of the following values N for no errors. R for motor running. O for overload error (motor has lost steps ; the syringe must be re- initialized I for a syringe not initialized or being initialized. M for a syringe motor missing ; either it is not installed (single syringe 402) or the syringe drive is defective or locked because of a software error H for an uncompleted motion; either the motion has been specified by a B A or D command but not started yet, or the syringe has been halted by a B H command. W for a syringe waiting for the other syringe as set by the B command W nnnnnn is the left syringe contents volume. The unit is the microliter, or the step if syringe 39000 is used g is the right syringe status, identical to f (see above). In the case of an 0 error, the syringe must be reinitialized before normal operation can resume.

VERITY® Syringe Pumps (v1.0.1.5)			402 Syringe Pumps (v2.32)		
Command	*Type*	*Description*	*Command*	*Type*	*Description*
P*n:v.vvv:s*	B	Sets syringe pump parameters: n = valve position; N for needle (probe) position, R for reservoir position, A for air/gas (3-way valve), or 1–5 for valve position (5-way valve) v.vvv = volume (in microliters) to aspirate (+) or dispense (-) s = (Optional) flow rate in mL/min. If flow rate is not specified, the default flow rate for the selected syringe will be used. Use immediate F to obtain that information. For example: Use buffered PN:-500:2 to dispense 500 µL at 2 mL/min to the needle (probe). Use buffered PR:2500:5 to aspirate 2500 µL (2.5 mL) at 5 mL/min from the reservoir. Use buffered PX to halt the syringe (emergency stop) and g error 12. Use buffered PS for a safe stop. The syringe operation will stop in a controlled manner and the position and held volume information is maintained. The instrument state is normal (no error).			
P	I	Returns ‘n:v.vv’, where: n is the valve position and may be: N, R for 2-Way valve position N, R, or A for 3-Way valve position 1-5 for 5-Way valve position V for valve not present or vacant. v.vvv is the syringe contents volume in unit of µL or ‘?’ if unknown(unhomed)			

VERITY® Syringe Pumps (v1.0.1.5)			402 Syringe Pumps (v2.32)		
Command	*Type*	*Description*	*Command*	*Type*	*Description*
This command has been replaced by the buffered @ command (address 4). Refer to the Non-Volatile Memory Map on page 13.			P	B	Set syringe size Syntax: Pnvvvvv where n is the syringe identification n = L for left syringe n = R for right syringe n = B for both syringes vvvvv is the syringe volume, in microliters. The syringe size for both syringes must be taken in the following list: 100, 250, 500, 1000, 5000, 10000, 25000. The value 39000 is also accepted to indicate that the commands A, D and S will be expressed in steps or steps/s.
p	B	Homes the syringe pump, which sends the piston to the upper position with the valve in the needle (probe) position. For a 5-way valve, the home position is user-settable using the buffered @ command (address 5). Refer to the Non-Volatile Memory Map on page 13. **Note**: Before sending this command, ensure that the probe is in a suitable location (rinse site, for example). This will help prevent the spilling of waste liquid or rinse diluent.			
Q	I	Reads the pressure sampling frequency. Displays in samples per second.			
Q	B	Sets the sampling frequency. Range is 20.00–0.01 samples per second.			
s	I	Service Query. Returns ‘Syringe = n’ or ‘Valve = n’ where n is the number of movements that the syringe has made or the number of movements that valve has made.			

<table>
<tr><th colspan="3">VERITY® Syringe Pumps (v1.0.1.5)</th><th colspan="3">402 Syringe Pumps (v2.32)</th></tr>
<tr><td>Command</td><td>Type</td><td>Description</td><td>Command</td><td>Type</td><td>Description</td></tr>
<tr><td colspan="3">There is not an equivalent command for the VERITY syringe pumps.</td><td>S</td><td>I</td><td>Return global status
Response format: ab where
a is the command status buffer
a = 0 for buffer empty
a = 1 for buffer occupied.
When the buffer is empty, the commands previously sent are being processed; when it is occupied, they are not necessarily being processed, and it is not possible to send more B commands.
b is the error flag status
b = 0 for no error, all commands have been accepted.
b = 1 for one or more rejected B commands since the last I $ command.
This flag is reset to 0 just after the immediate $ command has been executed.</td></tr>
<tr><td colspan="3">This command has been replaced by the buffered P command.</td><td>S</td><td>B</td><td>Set syringe flow
Syntax: Snvvvvv where
n is the syringe identification
n = L for left syringe
n = R for right syringe
vvvvv is the flow in ml/min, or in steps/s if the syringe 39000 is used.
<table>
<tr><td colspan="9">Volume range</td></tr>
<tr><td>Syringe (μl)</td><td>100</td><td>250</td><td>500</td><td>1000</td><td>10000</td><td>5000</td><td>25000</td><td>39000</td></tr>
<tr><td>Flow min (ml/min)</td><td>0.001</td><td>0.001</td><td>0.001</td><td>0.01</td><td>0.01</td><td>0.02</td><td>0.04</td><td>1*</td></tr>
<tr><td>Flow max (ml/min)*</td><td>6</td><td>15</td><td>30</td><td>60</td><td>120</td><td>240</td><td>240</td><td>39000*</td></tr>
<tr><td>Increment (ml/min)</td><td>0.001</td><td>0.001</td><td>0.001</td><td>0.01</td><td>0.01</td><td>0.01</td><td>0.01</td><td>1*</td></tr>
</table>
If the parameter flow is out of range, the flow rate will be set to either the maximum or the minimum possible. This event will be flagged by the I S command. The new syringe flow rate will be effective at the next aspirate or dispense command. The syringe flow cannot be changed while the syringe is running.</td></tr>
<tr><td>U</td><td>I</td><td>Reads the pressure sensor unit of measurement.</td><td></td><td></td><td></td></tr>
<tr><td>U</td><td>B</td><td>Sets the pressure sensor data format.
(P)SI = Pound-force per Square Inch
1 (B)AR = 14.5037738 PSI
1 (M)PA = 145.037738 PSI
(R)AW = Digital ADC output value</td><td></td><td></td><td></td></tr>
<tr><td>+</td><td>I</td><td>Returns ‘CRC = XXXX’ Where XXXX is the current value of the CRC.</td><td></td><td></td><td></td></tr>
</table>

VERITY® Syringe Pumps (v1.0.1.5)			402 Syringe Pumps (v2.32)		
Command	*Type*	*Description*	*Command*	*Type*	*Description*
This command has been replaced by the buffered P command.			B	B	Start syringe Syntax: Bn where n is the syringe identification n = L for left syringe n = R for right syringe n = B for both syringe Send the designated syringe(s) to the destination defined by the previous B A or D commands if any. If both syringes are selected, they start simultaneously. If the destination is not defined for a syringe, this command has no effect. The syringe(s) will not start until the valve on the same side (if any) is at rest. Furthermore, the start of the syringe can be delayed under more conditions, using the B command T.
This command has been replaced by the immediate P command.			V	I	Valve status Response format: ab where a is the left valve status b is the right valve status. R for valve to reservoir. N for valve to needle. X for valve running. O for valve overload error (the motor has lost steps) or failure to detect one or both rest positions. May happen on initialization. M for valve absent.

<table>
<tr><th colspan="3">VERITY® Syringe Pumps (v1.0.1.5)</th><th colspan="3">402 Syringe Pumps (v2.32)</th></tr>
<tr><td>Command</td><td>Type</td><td>Description</td><td>Command</td><td>Type</td><td>Description</td></tr>
<tr><td colspan="3">This command has been replaced by the buffered P command.</td><td>A</td><td>B</td><td>Aspirate volume
Syntax: Anvvvvv where
n is the syringe identification.
n = L for left syringe
n = R for right syringe.
vvvvv.v is the aspirated volume in microliters (0 up to syringe volume), or in steps if syringe 39000 is used.
This command sets the next aspiration but does not initiate liquid flow. To begin aspiration, use the B B command.
Volume range
<table>
<tr><td>Syringe (µl)</td><td>100</td><td>250</td><td>500</td><td>1000</td><td>5000</td><td>10000</td><td>25000</td><td>39000</td></tr>
<tr><td>Vol. min (ml)</td><td>0.1</td><td>0.1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0*</td></tr>
<tr><td>Vol. max (ml)</td><td>100</td><td>250</td><td>500</td><td>1000</td><td>5000</td><td>10000</td><td>25000</td><td>39000*</td></tr>
<tr><td>Increment (ml)</td><td>0.1</td><td>0.1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1*</td></tr>
</table>
This command will be rejected if the syringe involved has not been defined and successfully initialized or if it is an error. Use the I M command to see if there no error after initialization. If the volume is greater than the volume that can be aspirated by the syringe, the command will be rejected. Rejection is flagged by the I S command.
* These units are in steps, not uL.</td></tr>
<tr><td colspan="3">This command has been replaced by the buffered P command.</td><td>D</td><td>B</td><td>Dispense volume
Syntax: Dnvvvvv
n is the syringe identification
n = L for left syringe
n = R for right syringe
vvvvv is the dispensed volume in microliters (0 up to syringe volume), or in steps if syringe 39000 is used.
This command sets the next dispense but does not initiate liquid flow.
To begin dispense, use the buffered B command.
Volume range: see B A command above.
If the volume is greater than the volume currently held by the syringe, the command will be rejected.
This is flagged by the I S command. This command will be rejected until the syringe involved has been defined and successfully initialized or it is an error. Use the I M command to check if there is no error after initialization.</td></tr>
<tr><td colspan="3">This command has been replaced by the buffered P command.</td><td>N</td><td>B</td><td>Halt syringe motors
Syntax: Hn where
n is the syringe identification
n = L for left syringe
n = R for right syringe
n = B for both syringes
If the specified syringe is at rest, the command is ignored. No error is flagged.</td></tr>
</table>

VERITY® Syringe Pumps (v1.0.1.5)	402 Syringe Pumps (v2.32)		
Command / *Type* / *Description*	*Command*	*Type*	*Description*
This command has been replaced by the buffered p command.	O	B	Initialize syringe Syntax: On where n is the syringe identification n = L for left syringe n = R for right syringe n = B for both syringe This command sends the piston to the topmost position, until the plunger touches the valve body; then the piston moves down by a small amount to provide a clearance between the piston and the syringe top. The motor force is set automatically according to the syringe selected. If necessary the force may be adjusted with the B F command before initializing the syringe. Care must be taken that some liquid will be dispensed during initialization. If the syringe motor is missing, this command has no effect. This command acts immediately, even if a motion is already in progress. After successful initialization, the I M command returns N for no error and the contents of the syringe are set to zero. If the initialization failed, the M error is returned. The syringe must be initialized prior to any syringe movement command.
There is not an equivalent command for the VERITY syringe pumps.	T	B	Timely start Syntax: Tn where n is the syringe identification n = L for left syringe n = R for right syringe This command delays the motion of the specified syringe to synchronize it with the other syringe. It must be sent before the buffered B command, and may be sent before or after the B A or D command that defines the next motion. For a single syringe, single valve 402, it does not apply. For a dual syringe, single valve 402, the syringe waits until the other syringe and the valve are at rest. For a dual syringe, dual valve, the syringe waits both the other syringe and the other valve. Even if this command is not sent, the syringe will not start until the valve on the same side is at rest.
This command has been replaced by the buffered @ command (address 3). Refer to the Non-Volatile Memory Map on page 13.	U	B	Valve configuration option Syntax: Un where n is 1 or 2. For a 402 dilutor fitted with two valve drives, sending U1 forces the right-hand valve to be missing as though the motor had not been installed. This is necessary to allow a proper sequencing of the syringes in case the functions that are specific to a single valve, dual syringe configuration are used with a dual valve dilutor. Sending U2 or the immediate $ command restores the dual syringe configuration. This command has no effect in single-syringe, or dual-syringe and single valve configurations.

VERITY® Syringe Pumps (v1.0.1.5)			402 Syringe Pumps (v2.32)		
Command	*Type*	*Description*	*Command*	*Type*	*Description*
This command has been replaced by the buffered P command.			V	B	Valve control Syntax: Vnp where n is the syringe identification: n = L for left syringe n = R for right syringe p is the desired valve position: R for valve to reservoir. N for valve to needle. If the valve is already at the desired position or if the valve is missing, this command has no effect.

Non-Volatile Memory Map

VERITY Syringe Pumps

Address	Meaning	Units	Decimal Places	Default
0	Reserved			
1	nvBaud Sets the baud rate for GSIOC unit IDs above 9–63. Possible values are 19200; 57600; 115200	Baud	0	115200
2	nvNameID 0 = VERITY 4020 1 = VERITY 4120 2 = VERITY 4220 3 = VERITY 4060 4 = VERITY 4260 5 = Syringe Pump		0	0
3	Valve Type Contains integer valve type value for machines with 6 slot valve encoder. Permitted values: 0 = No Valve 2 = 2-Way Valve 3 Position 3 = 3-Way Valve 6 Position 5 = 5-Way Valve 6 Position		0	2
4	Syringe Size Contains syringe size in µL as an integer. Permitted values: 100 = 100 µL syringe 250 = 250 µL syringe 500 = 500 µL syringe 1000 = 1 mL syringe 5000 = 5 mL syringe 10000 = 10 mL syringe 25000 = 25 ml syringe	µL	0	10000
5	5-Way Valve Home Position Contains integer valve home (needle) position for machines with 6 slot valve encoder and 5-Way Valve. Permitted values: 1, 2, 3, 4, 5 Data is only used with the 5-Way Valve. Location is not modifiable unless 5-Way valve is selected in NV-RAM location 3.		0	1
6	Backlash	µSteps	0	60

Address	Meaning	Units	Decimal Places	Default
7	NVUnitID Sets the GSIOC unit ID.		0	11
8	Reserved			
9				
10–19	Reserved			
20	Offset valve		0	0
21	Offset syringe	µSteps	0	560
22–33	Reserved			
34	Pressure sensor Units 0 = Raw ADC counts 1 = PSI 2 = Bar 3 = MPa		0	1
35	Pressure sensor zero	Integer	0	0
36	Pressure sensor scale	Integer	0	0
37	Pressure sample frequency	Integer	0	10
38	Pressure sensor intercept	Integer	0	0
39	Pressure sensor slope	Integer	0	74008

Error Messages

VERITY Syringe Pumps

Error	Error Text	Error	Error Text
0	No Error	26	Invalid Valve Position
10	Unknown Command	27	Invalid Syringe Pump Volume
11	Invalid NV-RAM Address	28	Invalid Valve Move Speed
12	Emergency Stop Activated	29	Invalid Syringe Pump Flow Rate
13	Bad Parameter Entered	30	Invalid Valve Type
16	Character Limit	31	Invalid Syringe Size
18	Valve Park Location	32	Invalid Valve Selection
19	Syringe Pump Park Location	33	Invalid Syringe Pump Selection
20	Valve Unhomed	34	Missing Valve Encoder
21	Syringe Pump Unhomed	35	Missing Syringe Pump Encoder
22	Valve Moving	36	Set Pressure Offset to Zero
23	Syringe Pump Moving	37	Other Syringe Module in Error
24	Valve Stall		
25	Syringe Pump Stall		